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End-to-end stereoscopic video streaming with content-adaptive rate and format control

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Abstract

We address efficient compression and real-time streaming of stereoscopic video over the current Internet. We first propose content-adaptive stereo video coding (CA-SC), where additional coding gain, over that can be achieved by exploiting only inter-view correlations, is targeted by down-sampling one of the views spatially or temporally depending on the content, based on the well-known theory that the human visual system can perceive high frequencies in three-dimensional (3D) from the higher quality view. We also developed stereoscopic 3D video streaming server and clients by modifying available open source platforms, where each client can view the video in mono or stereo mode depending on its display capabilities. The performance of the end-to-end stereoscopic streaming system is demonstrated using subjective quality tests.
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Keywords: Stereoscopic video; Streamer; Content-adaptive; H.264

1. Introduction

The next big step forward in video technology is destined to be three-dimensional (3D) video display and distribution. While there are many alternative technologies for 3D video, including holographic, volumetric and stereoscopic, stereoscopic 3D video seems to be the most mature technology at the moment. Hence, this paper addresses efficient compression and streaming of stereoscopic video over the current Internet. Stereoscopic video consists of two video sequences (left and right views) captured by closely located (approximately the distance between two eyes) cameras; therefore, the bitrate requirement of a stereoscopic video will be higher than the monoscopic video. However, there are two factors that can be exploited for efficient encoding and streaming of multi-view video: (i) Inter-view redundancy: the close distance between the cameras results in high correlation between the two views. (ii) Psycho-visual redundancy: it is well

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known that the human visual system can perceive high frequencies in 3D from the higher quality two-dimensional (2D) view, if the two 2D views do not have the same spatial/temporal resolution [26].

There are many research and standardization activities for stereoscopic video compression based on exploiting inter-view redundancy. Early work in this area resulted in the MPEG-2 multi-view profile [20]. Later [18] proposes modifications to MPEG-2 multi-view profile for improving the compression efficiency again based on the correlation between two views. Recently, new stereoscopic video codecs based on H.264 are introduced in [1,2,15]. A new standard for multi-view video coding (MVC) is currently under development under the auspices of Joint Video Team (JVT) [17]. This paper proposes making use of the psycho-visual redundancy depending on the characteristics of the video content in order to achieve additional compression efficiency.

In monoscopic video compression, it is common practice to sub-sample the chrominance channels, since the human visual system (HVS) is less sensitive to variations in chrominance values. Similarly, in the theory of stereo perception, it is reported that the HVS can perceive high frequency information in 3D from one of the views even if the other view is low pass filtered [26]. Hence, spatial and temporal subsampling can be performed to reduce the bandwidth requirements. To this effect, we propose a content adaptive approach for temporal and spatial downsampling of one of the views to achieve better compression with higher perceptual quality in our H.264 based multi-view codec.

We also propose an end-to-end stereoscopic 3D video streaming architecture using the proposed content-adaptive multi-view coding and modifications to available open source monocular streaming platforms. There are several open source monocular video streaming platforms, including the Darwin Streaming Server (DSS) [8], GPAC [9] and VideoLAN Client (VLC)/Server [10]. Apple QuickTime Streaming Server (QSS) and its open source version (DSS) support streaming of H.264 [13] coded video wrapped inside MPEG-4 [12] or 3GPP file format across the Internet using RTSP and RTP protocols [24]. In order to stream media over RTP, these systems need special information about the media files, which is carried in a hint track. Another project called GPAC, which is developed as a multimedia framework based on MPEG-4 standard, also supports streaming H.264 coded media files inside MPEG-4 file format [9]. In addition to these two platforms, VLC also provides streaming capabilities. VLC supports H.264 video in local playback and streaming when encapsulated in MPEG-TS file format over RTP. Recently, a 3DTV prototype system, similar to our system, with real-time acquisition, transmission and auto-stereoscopic display of dynamic scenes, has been offered by MERL. Multiple video streams are encoded and sent over a broadband network. The 3D display shows high-resolution stereoscopic color images for multiple viewpoints without special glasses. This system uses light-field rendering to synthesize views at the correct virtual camera positions [28].

The rest of the paper is organized as follows: In Section 2, we present the proposed content-adaptive multi-view video encoding. Section 3 describes the end-to-end streaming system in detail. Results are given in Section 4, and Section 5 provides the conclusions.

2. Content-adaptive stereo video coding

There are conflicting theories about the effects of unequal distortion in the left and right video sequences, e.g., the fusion theory and suppression theory [5,14,30]. According to the fusion theory, the stereo bitrate (hence distortion) needs to be equally allocated between the views for the best human perception. Contrarily, according to the suppression theory, the highest quality view determines the overall perception performance in stereo-video. Therefore, the target (e.g., right) sequence can be compressed more to save more bits for the reference (e.g., left) sequence, such that the overall perceived distortion is the lowest. The proposed content-adaptive stereo codec (CA-SC) is motivated by the suppression theory and reduces the spatial resolution and/or the frame (temporal) rate of the target (right) sequence adaptively according to its content-based features.

The principle behind content adaptive video coding is to parse video into temporal segments. Each temporal segment can be encoded at different spatial, temporal and SNR resolutions (hence at a different target bitrate) depending on its low and/or high-level content-based features. Even though this approach has been used for monoscopic video encoding [4,16,19,27], there are no such studies in the literature for content-adaptive stereoscopic coding. The proposed CA-SC is an extension of

the stereo codec (SC) in [2] which is based on AVC/H.264. We note that CA-SC can also be developed as an extension of the recently standardized MVC codec [17]. In stereoscopic coding, in the compatible mode, any standard H.264 decoder can decode the sequence as a monoscopic sequence since left channel is coded independent of the right channel. In order to improve the coding efficiency without significant perceptual quality loss, we added three modes to the encoder for downsampling the right-view only: they are the spatial, temporal and content-adaptive scaling modes.

2.1. Spatial scaling

The spatial scaling mode corresponds to downsampling the right view by a predefined scale prior to encoding. The implementation of downsampling consists of both decimation and low-pass filtering in order to prevent aliasing. For spatial scaling the following filters are used:

13-tap downsampling filter:
$\{0, 2, 0, -4, -3, 5, 19, 26, 19, 5, -3, -4, 0, 2, 0\}/64$,
11-tap upsampling filter:
$\{1, 0, -5, 0, 20, 32, 20, 0, -5, 0, 1\}/64$.

Filters are applied to all Y, U and V channels and in both horizontal and vertical directions. The picture boundaries are padded by repeating the edge samples. These filters are currently used in Scalable Video Coding extension of H.264 [22] and explained in [25]. In order to keep filtering process simpler in both encoder and decoder, we have implemented downscaling by factors of 2 (dyadic sampling) in both dimensions. Although the spatial scaling is applied to the right view only in our stereoscopic codec, during the motion estimation left frames are also scaled the same amount for proper estimation.

2.2. Temporal scaling

Temporal scaling mode corresponds to the decimation of the right view in time, i.e., frame dropping in the right sequence. The implementation of temporal downsampling is done by sending all macro-blocks of dropped frames as skip mode of the H.264 standard. In our codec notation, temporal scaling of n denotes encoding 1 frame out of n frames and dropping the remaining $n-1$ frames.

2.3. Content-adaptive scaling

In content-adaptive video scaling, we first divide the right video into temporal segments (shots or sub-shots) using well-known temporal segmentation methods [31]. We then classify the temporal segments (shots) into four categories as determined by their low-level attributes such as the amount of motion and spatial detail within the segment. Shots with high temporal activity (high motion) need to be encoded at full temporal resolution for a smooth viewing experience. On the other hand, if a somewhat stationary shot is being encoded, the temporal sampling rate can be reduced to a lower value without any loss of perceptual quality. Likewise, shots with high spatial detail should not be reduced to lower spatial resolutions for the sake of perceptual quality, while it is harmless to do such downsampling in case of low spatial detail.

There have been several studies on temporal content adaptation and classification in the literature over the past years [6,31], which can be used to determine which temporal region belongs to what spatial and temporal class in any given video segment. The spatio-temporal attributes may vary even from GOP to GOP (Group of Pictures, i.e., each chunk of video sequence that can be decoded without use of other parts of the sequence) within a shot. In the following, we define *spatial scene complexity* and *temporal activity* measures for the classification of temporal segments.

The *spatial scene complexity measure* is calculated as the pixel variance across the temporal segment.

Assume that the number of frames (either left or right sequence) in a specific stereoscopic video temporal segment is N. Let the pixel resolution of each frame be $H \times W$ where the symbol H denotes the height and the symbol W denotes the width of the frames in the number of pixels. In order to calculate the spatial complexity measure of a single frame i, we first find the mean of horizontal and vertical squared pixel value differences as follows:

$$E[d_i^2] = \frac{\left(\sum_{h=0}^{H-2}\sum_{w=0}^{W-1}\{(p_i(h+1,w)-p_i(h,w))^2\} + \sum_{h=0}^{H-1}\sum_{w=0}^{W-2}\{(p_i(h,w+1)-p_i(h,w))^2\}\right)}{W(H-1)+H(W-1)},$$

where h represents the vertical pixel coordinates and w represents the horizontal pixel coordinates within the frame. We then calculate the square of the mean

pixel difference as indicated by

$$E[d_i]^2 = \left(\frac{\left(\sum_{h=0}^{H-2} \sum_{w=0}^{W-1} (p_i(h+1,w) - p_i(h,w)) + \sum_{h=0}^{H-1} \sum_{w=0}^{W-2} (p_i(h,w+1) - p_i(h,w)) \right)}{W(H-1) + H(W-1)} \right)^2.$$

Therefore, the pixel variance within the frame can be calculated as follows:

$$\sigma_i^2 = E[d_i^2] - E[d_i]^2.$$

Hence the scene complexity, σ, of the temporal segment can be calculated as the average pixel variance across its frames:

$$\sigma = \frac{1}{N} \sum_{i=0}^{N-1} \sigma_i^2.$$

A segment is classified as having low spatial scene complexity if this measure is below some threshold, and as having high spatial scene complexity otherwise. The *temporal activity measure* is calculated for each frame by taking the average absolute motion vector length (encoder output) in a temporal segment. The intra-block codings within inter-coded frames (P or B frames) are considered to have the maximum motion vector length within the allowable search range. The intra-coded I-frames are left out of this temporal activity measure computation. After this, the results found are scaled by their respective spatial resolutions in order to be consistent across different stereoscopic videos. A shot is classified as having low temporal activity, if this measure is below a threshold; and classified as high temporal activity if it exceeds the threshold. Therefore, one can classify temporal video segments into four categories with specific low-level temporal and spatial attributes, where the appropriate spatial and temporal formats for the right view only belonging to each class are as follows:

Type 1: High spatial and temporal activity: do not scale the spatial and temporal formats.
Type 2: Low spatial and high temporal activity: apply spatial scaling but not temporal scaling.
Type 3: High spatial and low temporal activity: apply temporal scaling but not spatial scaling.
Type 4: Low spatial and temporal activity: apply both temporal and spatial scaling.

3. End-to-end stereo video streaming system overview

An overview of the proposed system architecture is shown in Fig. 1. The stereoscopic display system which is composed of a pair of projectors can be seen in Fig. 2. Stereo video is encoded off-line by using the proposed CA-SC, which has been described in Section 2. The basic codec structure of the CA-SC can be seen in Fig. 3. The server, which is detailed in Section 3.1, streams the encoded bitstream over the Internet. The classification of streamed video in the content adaptive codec is illustrated in Fig. 4. The end users can view either monoscopic or stereo streams based on their display capabilities using the client, described in Section 3.2.

3.1. Server

The server employs the protocol stack RTP/UDP/IP, and can serve multiple clients, described in Section 3.2, simultaneously. Session description protocol (SDP) is used to ensure interoperability with the clients.

Since the proposed CA-SC encoder is a modification of the H.264/MPEG-4 AVC design, it supports a Video Coding Layer (VCL), which efficiently represents the video content, and a Network Abstraction Layer (NAL), which provides header information for particular transport layers (such as real-time transport protocol, RTP) or storage media. All data are contained in NAL units, each of which contains an integer number of bytes. The format of NAL units is the same for both packet-oriented transport and bitstream delivery. The only difference is each NAL unit can be preceded by a start code prefix in a bitstream-oriented transport layer [13].

The encoded stereo bitstream contains NAL units of both left and right views. These NAL units are packetized for independent streaming over two separate channels using RTP [9]. The sender side packet format is implemented based on the RTP payload format for H.264 video [29]. Three packetization modes are defined in this payload format. We implemented Single NAL Unit mode and Non-Interleaved mode, which are intended for low-delay applications. We used FU-As (Fragmentation Unit without Decoding Order Number) packetization structure to transfer NALUs the sizes of which are

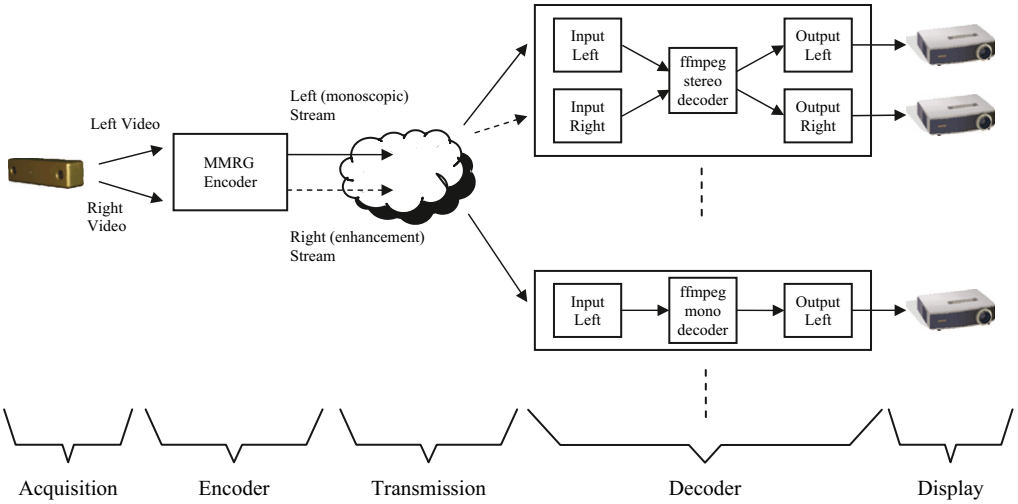

Fig. 1. End-to-end system overview.

exceeding the network MTU. We fragmented the NALUs on the application layer instead of relying on the IP layer fragmentation. Other packets with smaller sizes are sent in Single NAL unit packets.

In both of these packetization modes, the transmission order of the RTP packets shown by the sequence numbers is taken as the decoding order of the NAL units. Since our encoder does not support B frames, packet structures, which do not contain decoding order numbers are usable in our application. Timestamps carried in the RTP header are used to determine the decoding order of the frames. The RTP timestamps are used to synchronize the frames. The display application arranges the play-out time by using the relative order of the frames positioned by the RTP timestamps. Since we stream two video files, we set related frames to the same timestamp supposing same sampling rate for videos with a 90 kHz clock. In addition, the H.264 parameter sets are fundamental parts for video coding. A more reliable transfer is required for their transmission and receiver must receive them before the decoding process. So we transfer them out-of-band to the receiver side reliably prior to the actual RTP sessions.

For the interoperability of the stereo video server and the client on the receiver side, we used SDP [7]. For indicating stereo view, an additional session attribute is used in order to specify stereo data and which channel is the left and which one is the right. Moreover, for future extension of stereo streaming to multi-view streaming, the session descriptor also can be used. Currently, we define a new attribute "view" which gives the address and the port information of the other sessions broadcasting extra views of the video.

a = view : mono	
a = view : stereo	⟨*address-Left*⟩ ⟨*port-Left*⟩ ⟨*address-Right*⟩ ⟨*port-Right*⟩
a = view : multi	⟨*address*⟩ ⟨*port*⟩ ⟨*address*⟩ ⟨*port*⟩, ⟨*address*⟩ ⟨*port*⟩ ⟨*address*⟩ ⟨*port*⟩, ⟨*address*⟩ ⟨*port*⟩ ⟨*address*⟩ ⟨*port*⟩, …

where "mono" for monoscopic, "stereo" for stereoscopic and "multi" for multi-view gives the view type and "⟨address⟩ ⟨port⟩ ⟨address⟩ ⟨port⟩ "pair gives the access information of two corresponding views of the multi-view video.

3.2. Clients

We have implemented three clients for different types of display systems: (i) Client-1 supports an in-house polarized 3D projection display system; (ii) Client-2 supports the auto-stereoscopic Sharp 3D laptop; (iii) Client-3 supports a monocular display to demonstrate backwards compatibility. For all client implementations, we modified the open source software VLC. VLC is a highly portable multimedia player for various audio and video formats and streaming protocols [10]. We modified its stream receiver to support raw H.264 streaming over RTP

and used it as a player. The modified VLC handles packets of left and right views using two separate threads. We send NALU units received in RTP packet payload directly to the decoder after de-packetization.

As the decoder, open source H.264 decoder implementation in the FFmpeg library [11] is used with MVC modifications. Before sending to the decoder, the data are buffered in order to synchronize related left and right frames. The decoder decodes and sends the decoded picture to the video output modules. The video output units visualize the left and right frames in a synchronized manner by using the time information in the RTP timestamps.

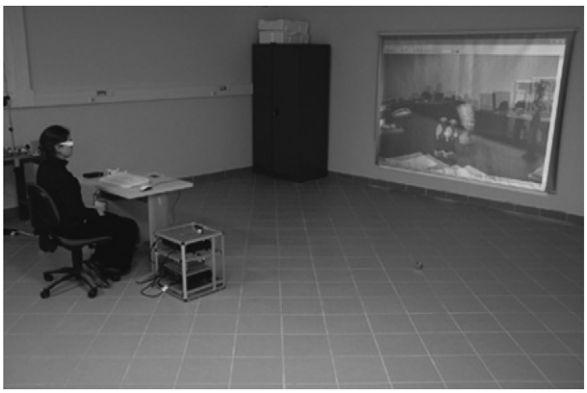

Fig. 2. Stereoscopic display system.

Finally, the in-house 3D projection display system uses a pair of Sharp MB-70X projectors as shown in Fig. 2. Light from one of the projectors is polarized in clockwise direction and light from the other projector in counter-clockwise direction using circular polarization filters. Both projectors are precisely aligned to project onto a special silver screen covered with a neutral grey reflective dielectric material to preserve the polarization of light during reflection. The users wear glasses, which have matching filters with the projectors to ensure that light from one projector is only seen with one eye. A single high-end PC with two display outputs using the extended desktop feature drives the projectors. This setup results in a virtual desktop of 2048×768 pixels, each projector displaying only one half of the extended desktop at 1024×768 native resolutions. Thus, left and right videos can be easily shown on the left and right halves of the extended desktop, such that they exactly overlap with each other on the silver screen (Figs. 3 and 4).

4. Experimental results

4.1. Subjective quality tests

In order to investigate the effects of spatial, temporal and content-adaptive scaling in stereoscopic

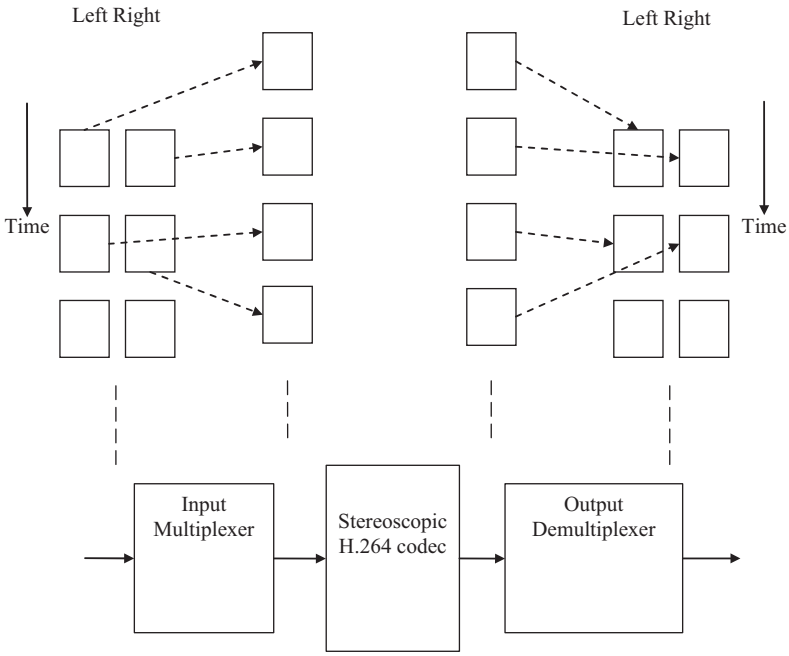

Fig. 3. Stereoscopic encoder.

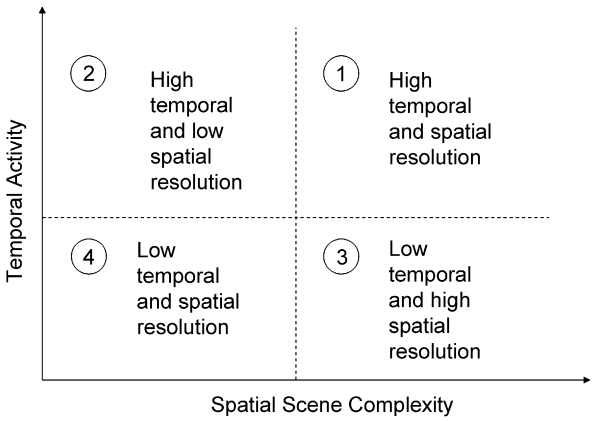

Fig. 4. Temporal and spatial formats appropriate for the right view according to low-level features.

videos, we employ the Double Stimulus Continuous Quality Scale (DSCQS) Method [21]. In this test method, assessors which are chosen among non-experts and inexperienced assessors should be used. The evaluation should be on a continuous scale ranging from 0 to 100. The method can be applied in two variants:

Variant 1: Each assessor is let to switch between two conditions, A and B (two stereoscopic images or videos), one of which is always the source and the other is the tested condition applied on the source. The identity of the images, whether it is the source or the test condition, should be known by the experimenter but not by the assessors. After evaluating the conditions the assessor moves to the next pair of images or videos.

Variant 2: Multiple assessors are shown two conditions, A and B (two stereoscopic images), consecutively one of which is always the source and the other is the tested condition applied on the source. The identity of the images, whether it is the source or the test condition, should be known by the experimenter but not by the assessors. The next pair of conditions is shown after the assessors establish an opinion.

Analysis method: For the analysis of the test results, each evaluation is graded between 0 and 100 and the difference between the scores of source image and the test condition is calculated to find the score of that test condition on that image by the assessor. After all these scores are calculated, the values are normalized to fit in 0–100. And as a final step, to find the scores of each algorithm (test condition) the average of all the scores over the assessors and images are taken. Scores of the algorithms can be compared with their closeness to the number to which zero score is mapped during the normalization process.

4.2. Experiments

In the experiments, we investigated effects of spatial, temporal and content-adaptive scaling in stereoscopic videos. In order to meet time requirements of assessment test, we use only four video sets with eight algorithms.

Assessors: Twenty-one assessors (13 females, 8 males with average age 24) with ages ranging from 19 to 36, volunteered to participate in the experiment. The participants were non-experts in the area of picture quality and were screened for color vision, stereo depth perception and visual acuity.

Each assessor is well informed on the test process and test materials (possible quality defects) before the test and they are assisted during the whole test procedure. DSQCS test method with the second variant mentioned above is used as the test methodology. At each step two video sequences, original left and right videos and processed left and right videos are used. We will call those four videos an evaluation pair. In the experiments, original videos are also repeated as a processed video in order to test the performance of the test.

At the beginning of the test, five random evaluation pairs are shown to the assessors and these five evaluation pairs are not evaluated since they provide stabilization of the perception of assessors. The test material is shown in a random order for each assessor. The randomization is done both among evaluation pairs and among the set of video sequences in the pair.

Test material: As the test material, four different stereoscopic video pairs are used: balloons (720×480, 25 fps, 10 s), botanical (960×540, 15 fps, 5 s), flowerpot (720×480, 25 fps, 10 s) and train tunnel (720×576, 25 fps, 10 s). The temporal activity values for each frame of the tested videos can be seen in Figs. 5–8 whereas the spatial scene complexity values are illustrated in Fig. 9. Eight different algorithms are applied on these videos as shown in Table 1. The algorithm ADAP is tested only on the balloons sequence since its motion vs pixel variance values of GOPs are the most scattered

among the other test sequences as in Fig. 13. As a result a total of 42 evaluation pairs, including first five stabilizing pairs, are shown to the assessors and it is assured that each test does not take more than 30 min.

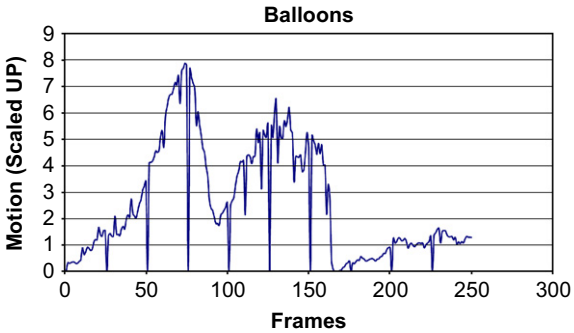

Fig. 5. The temporal activity values for balloons sequence.

4.3. Results

All the test videos are encoded with the modes explained in Table 1. Intra-period of 25 and Quantization Parameter (QP) of 28 are used while encoding. Total bitrate for simulcast (SIMUL) coding is interpreted as twice the data required compared to single view coding and the bitrates of all other algorithms are normalized accordingly and can be found in Table 2. The resulting average PSNR values of the sequences for different algorithms can be seen in Fig. 10. PSNR values are all in dB and calculated according to the following formulas where D_l and D_r

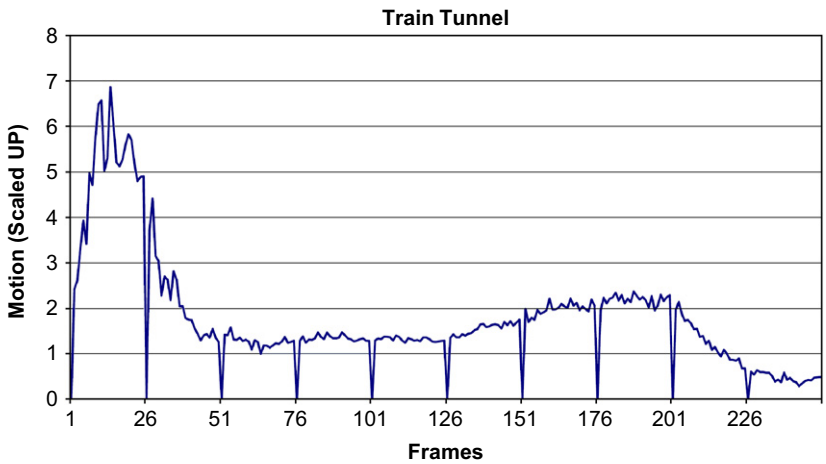

Fig. 6. The temporal activity values for train tunnel sequence.

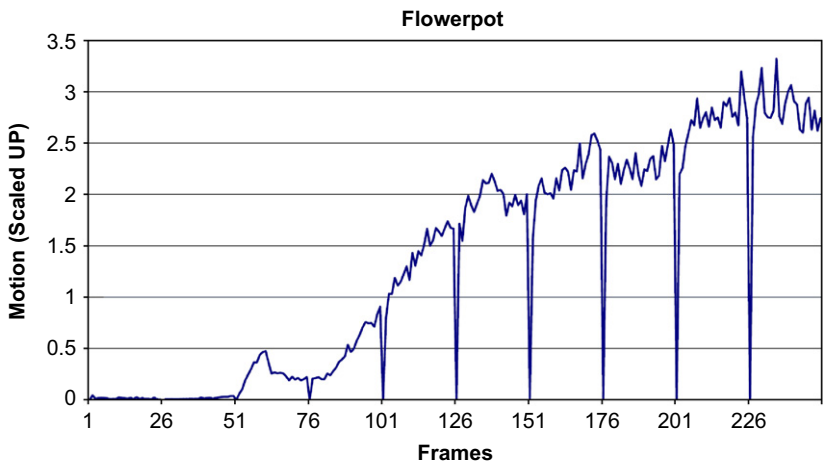

Fig. 7. The temporal activity values for flowerpot sequence.

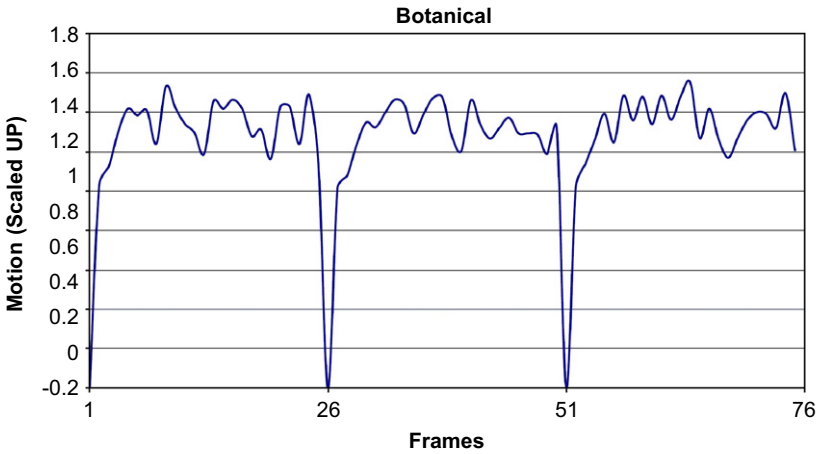

Fig. 8. The temporal activity values for botanical sequence.

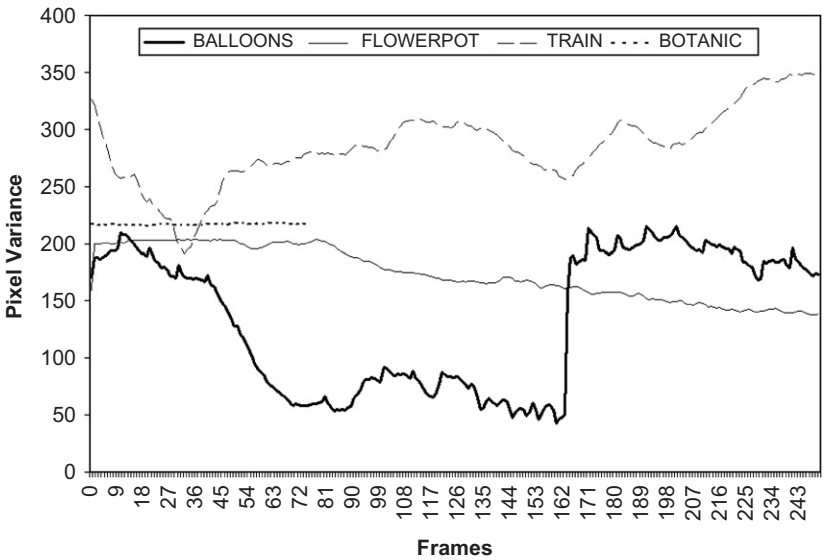

Fig. 9. Pixel variance (spatial scene complexity) values of each frame of the test videos.

Table 1
Algorithms applied to test videos

SIMUL	Simulcast coding
S1T1	Stereo coding, no spatial, no temporal scaling
S1T2	Stereo coding, no spatial, temporal scaling 2 for right frames
S1T2L	Stereo coding, no spatial, temporal scaling 2 for left and right frames
S1T3	Stereo coding, no spatial, temporal scaling 3 for right frames
S2T1	Stereo coding, spatial scaling 2, no temporal scaling
S4T1	Stereo coding, spatial scaling 4, no temporal scaling
S4T3	Stereo coding, spatial scaling 4, temporal scaling 3 for right frames
ADAP	Content-adaptive scaling algorithm

Table 2
Normalized bitrates of algorithms

	Baln	Flow	Bota	Train	Avrg.
SIMUL	2.000	2.000	2.000	2.000	2.000
S1T1	1.901	1.927	1.452	1.881	1.790
S1T2	1.606	1.692	1.289	1.601	1.547
S1T3	1.489	1.586	1.228	1.492	1.449
S1T2L	1.324	1.450	0.923	1.336	1.258
S2T1	1.242	1.267	1.065	1.252	1.207
ADAP	1.190	–	–	–	–
S4T1	1.091	1.095	1.012	1.085	1.071
S4T3	1.053	1.069	1.006	1.049	1.044

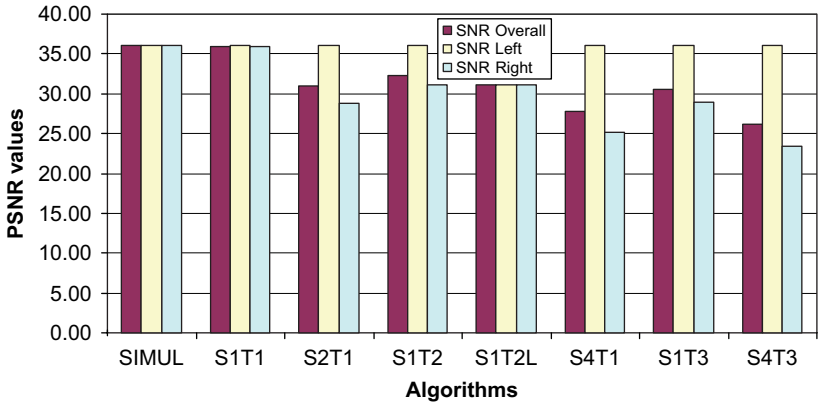

Fig. 10. PSNR values of the algorithms.

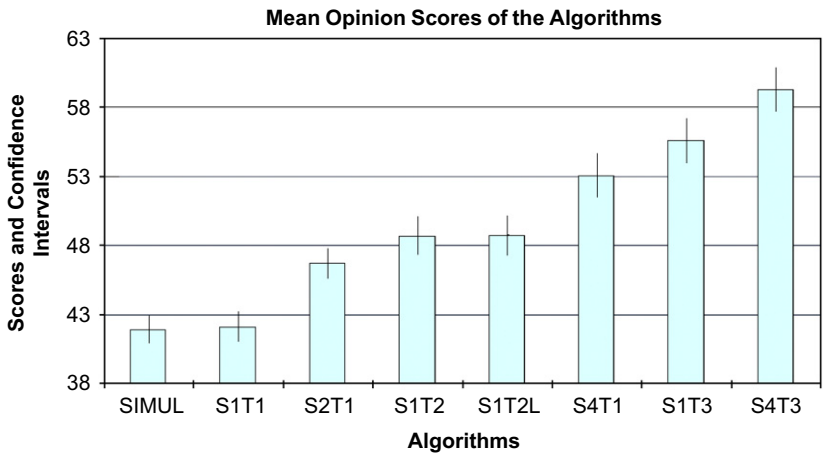

Fig. 11. Mean opinion scores and confidence intervals of the algorithms.

represent the distortions in right and left images [3]:

$$\mathrm{PSNR_l} = 10\log_{10}\frac{255^2}{D_\mathrm{l}/2},\quad \mathrm{PSNR_r} = 10\log_{10}\frac{255^2}{D_\mathrm{r}/2},$$

$$\mathrm{PSNR_{all}} = 10\log_{10}\frac{255^2}{(D_\mathrm{l}+D_\mathrm{r})/2}.$$

By only spatial subsampling of right video with 2 in both dimensions we approximately match 1.2 times the single view bitrate. By applying both spatial and temporal scaling on right frames, we can nearly code the stereoscopic video at single view bitrates.

After all the assessors finish the test, the scores are evaluated and normalized according to [21]. Average mean opinion scores (MOS) for each algorithm and confidence intervals are shown in Fig. 11. Due to the normalization, 0 (best quality) is mapped to 38, and the success of the algorithms can be measured by closeness of their mean to 38. This mapping is due to wrong evaluations of the assessors giving better scores to the distorted sequences than the original and expected in general. SIMUL coding and stereo coding without scaling (S1T1) have similar or better performances over original video. Since QP is low, reconstructed video quality is visually lossless (with average PSNR of 36 dB) and misjudgment is expected for these algorithms as well. Also DCT based coded images are reported [23] to be preferred by assessors compared to the original.

We can see that scaling with 3 or 4 in both spatial and temporal domains, results are not acceptable. Spatial scaling by 4 corresponds to 16:1 reduction in image size; therefore its performance is not acceptable. Spatial scaling with non-dyadic factors and

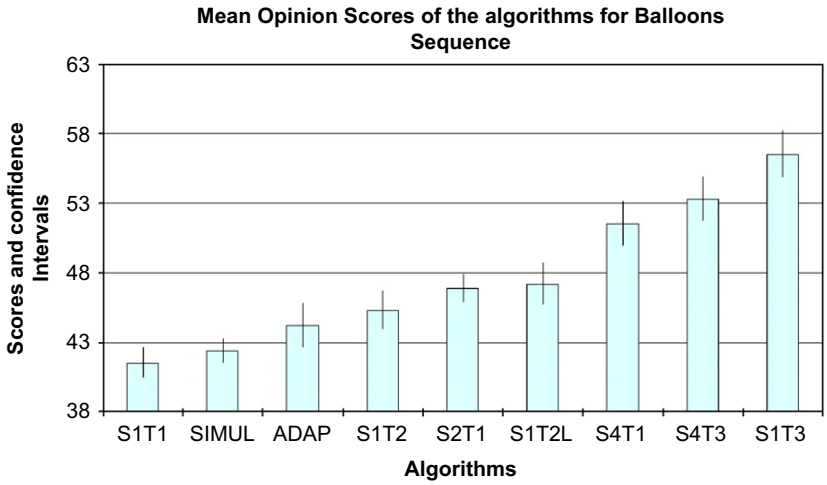

Fig. 12. Mean opinion scores and confidence intervals of the algorithms including the content-adaptive scaling algorithm for balloons sequence.

better filters for upsampling might keep the visual quality at desired levels with bitrate similar to single view coding bitrates.

According to the video characteristics (slow motion video), temporal scaling in either right channel or both channels gives better results (Fig. 12). By analyzing the characteristics of the video in each GOP, appropriate scaling is applied to decrease bitrate without visual quality degradation. The temporal (motion) and spatial (pixel variance) features of the GOPs of the test sequences can be seen in Fig. 13.

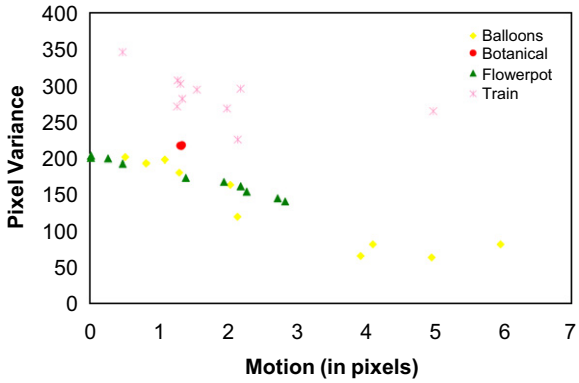

Fig. 13. Motion vs pixel variance averages of each GOP for each test sequence.

4.4. Streaming system performance

For transmission and display process, we have implemented all the modules and run the system with already encoded files. In order to cope with packet losses, frequent intra-frames are inserted and frames are coded in slice mode. Although the system is tested by encoding each frame as a single slice, number of slices can be increased or fixed size slices can be used according to the network state. The system is initially tried on local area network with zero packet losses. H.264 packet loss resilience techniques and loss concealment techniques will be added to the system for real Internet use.

The H.264 coded video increased the efficiency of bandwidth usage and this also affects the quality of the views. We coded videos as 25 fps and we inserted one intra-frame per 12 frame. Two different quantization values were used, one has Y channel limit value of 38.27 dB and the other one has 33.47 dB. For 320×240 video, their bandwidth usages were 744.665 and 415.335 kbits/s, respectively.

The system scalability was also another factor for the system design. The data can be multi-casted from anywhere and the users can view as mono or stereo depending on their connection capacity and display system. Moreover the player functionality and integrity also increase the usage of system with the future improvements on different file formats and codec standards.

5. Conclusions and future work

In this study we have implemented an end-to-end stereoscopic video streaming system using content-adaptive multi-view coding and modifications to available open source monocular streaming platforms.

Our proposed content-adaptive approach for temporal and spatial downsampling of one of the views yields better compression with higher perceptual quality. The performance of the proposed approach is tested on several stereo sequences using subjective quality tests. The system is initially tried on local area network with zero packet losses. H.264 packet loss resilience techniques and loss concealment techniques will be added to the system for real Internet use.

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References

[1] B. Balasubramaniyam, E. Edirisinghe, H. Bez, An extended H.264 CODEC for stereoscopic video coding, in: Proceedings of SPIE, 2004.
[2] C. Bilen, A. Aksay, G. B.Akar, A multi-view video codec based on H.264, ICIP 2006, October 2006.
[3] N.V. Boulgouris, M.G. Strintzis, A family of wavelet-based stereo image coders, IEEE Trans. Circuits Syst. Video Technol. 12 (10) (October 2002) 898–903.
[4] S.F. Chang, D. Zhong, R. Kumar, Real-time content-based adaptive streaming of sports video, IEEE Workshop on Content-Based Access to Video/Image Library, Hawaii, December 2001, pp. 139–146 (also United States Patent Application No. 20040125877).
[5] I. Dinstein, M.G. Kim, A. Henik, J. Tzelgov, Compression of stereo images using subsampling transform coding, Opt. Eng. 30 (9) (September 1991) 1359–1364.
[6] N.D. Doulamis, A.D. Doulamis, Y.S. Avrithis, K.S. Ntalianis, S.D. Kollias, An optimal framework for summarization of stereoscopic video sequences, in: Proceedings of the International Workshop on Synthetic—Natural Hybrid Coding and Three Dimensional Imaging, Santorini, Greece, 1999.
[7] M. Handley, V. Jacobson, SDP: session description protocol, RFC 2327, April 1998.
[8] http://www.apple.com/.
[9] http://gpac.sourceforge.net/.
[10] http://www.videolan.org/vlc/.
[11] http://ffmpeg.sourceforge.net/.
[12] ISO/IEC International Standard 14496 (MPEG-4), Information technology—coding of audio-visual objects, January 2000.
[13] ITU-T ISO/IEC 14496-10, Recommendation H.264: advanced video coding for generic audiovisual services, May 2003.
[14] B. Julesz, Foundations of Cyclopeon Perception, University of Chicago Press, Chicago, 1971.
[15] H. Merkle, T. Schwarz, A. Hinz, Smolic, T. Wiegand, Multiview video coding based on H.264/AVC using hierarchical B-frames, in: Proceedings of PCS 2006, Picture Coding Symposium, Beijing, China, April 2006.
[16] R. Mohan, J.R. Smith, S. Chung, Adapting multimedia Internet content for universal access, IEEE Trans. Multimedia 1 (1) (March 1999) 104–114.
[17] K. Müller, P. Merkle, A. Smolic, T. Wiegand, Multiview coding using AVC, MPEG Meeting—ISO/IEC JTC1/SC29/WG11, Bangkok, Thailand, MPEG06/M12945, January 2006.
[18] Oh, et al., Scalable stereo video coding for heterogeneous environments, MIPS 2004, Lecture Notes Comput. Sci. 3311 (2004) 72–83.
[19] T. Ozcelebi, A.M. Tekalp, M.R. Civanlar, Delay-distortion optimization for content-adaptive video streaming, IEEE Trans. Multimedia, 2007, to appear.
[20] A. Puri, R.V. Kollarits, B.G. Haskell, Basics of stereoscopic video, new compression results with MPEG-2 and a proposal for MPEG-4, Signal Process.: Image Commun. 10 (1997) 201–234.
[21] Rec. ITU-R BT.500-11, Methodology for the subjective assessment of the quality of television pictures, 2002.
[22] J. Reichel, H. Schwarz, M. Wien, Scalable video coding—Working Draft 3, JVT-R201, Bangkok, Thailand, January 2006.
[23] A. Schertz, Source coding of stereoscopic television pictures, in: Proceedings Of IEE International Conference on Image Processing and its Applications, Maastricht, The Netherlands, April 7–9, 1992, pp. 462–464.
[24] H. Schulzrinne , S. Casner, R. Frederick, V. Jacobson, RTP: a transport protocol for real-time applications, RFC 3550, July 2003.
[25] C.A. Segall, Study of upsampling/down-sampling for spatial scalability, JVT-Q083, Nice, FR, PL, 14–21 October 2005.
[26] L.B. Stelmach, W.J. Tam, D. Meegan, A. Vincent, Stereo image quality: effects of mixed spatio-temporal resolution, IEEE Trans. Circuits Syst. Video Technol. 10 (2) (2000) 188–193.
[27] P. van Beek, J.R. Smith, T. Ebrahimi, T. Suzuki, J. Askelof, Metadata-driven multimedia access, IEEE Signal Process. Mag. 20 (2) (March 2003) 40–52.
[28] A. Vetro, W. Matusik, H. Pfister, J. Xin, Coding approaches for end-to-end 3D TV systems, Picture Coding Symposium (PCS), December 2004.
[29] S. Wenger, M.M. Hannuksela, T. Stockhemmer, M. Westerlund, D. Singer, RTP payload format for H.264 video, RFC 3984, February 2005.
[30] W. Woo, A. Ortega, Optimal blockwise dependent quantization for stereo image coding, IEEE Trans. Circuits Syst. Video Technol. 9 (September 1999) 861–867.
[31] B.L. Yeo, B. Liu, Rapid scene analysis on compressed videos, IEEE Trans. Circuits Syst. Video Technol. 5 (December 1995) 533–544.